SymPy Bug Report

1 Bug 43: Collapsed Range for $\cos(x)/\cot(x)$

1.1 Minimal Reproducer

```
from sympy import S, cos, cot, pi, sqrt, symbols
from sympy.calculus.util import continuous_domain, function_range

x = symbols("x")
expr = cos(x)/cot(x)

dom = continuous_domain(expr, x, S.Reals)
rng = function_range(expr, x, S.Reals)

print("continuous_domain(cos(x)/cot(x), x, S.Reals) =", dom)
print("function_range(cos(x)/cot(x), x, S.Reals) =", rng)
print("value at pi/3 =", expr.subs(x, pi/3))
print("range contains sqrt(3)/2 =", rng.contains(sqrt(3)/2))
print("value at pi/2 =", expr.subs(x, pi/2))
```

1.2 Observed Behavior

```
SymPy version: 1.14.0
continuous_domain(cos(x)/cot(x), x, S.Reals) = Reals
function_range(cos(x)/cot(x), x, S.Reals) = {0}
value at pi/3 = sqrt(3)/2
range contains sqrt(3)/2 = False
value at pi/2 = nan
```

SymPy reports that the real domain is all of $\mathbb{R}$ and that the range is the singleton $\{0\}$. Both conclusions are inconsistent with direct substitution: $x = \pi/3$ is a defined point with value $\sqrt{3}/2$, while $x = \pi/2$ is not a defined point of the quotient.

1.3 Expected Behavior

The real domain should exclude points where the quotient is undefined, in particular the zeros of $\cot(x)$ and the poles of $\cot(x)$. The returned range should include nonzero values attained at defined points, including $\sqrt{3}/2$. More precisely, on its natural real domain,

$$\frac{\cos x}{\cot x} = \sin x$$

where the quotient is defined, but the original quotient is undefined at multiples of $\pi/2$. Its range is therefore

$$(-1, 1),$$

not $\{0\}$.

1.4 Mathematical Explanation

Using $\cot x = \cos x/\sin x$, the quotient satisfies

$$\frac{\cos x}{\cot x} = \frac{\cos x}{\cos x/\sin x} = \sin x$$

only at points where the original expression is defined. The original expression has denominator $\cot x$, so all zeros of $\cot x$, namely $x = \pi/2 + k\pi$, must be excluded. The expression also inherits undefined points from $\cot x$, namely $x = k\pi$. Thus the natural real domain is

$$\mathbb{R} \setminus \left\{ \frac{k\pi}{2} : k \in \mathbb{Z} \right\}.$$

At the defined point $x = \pi/3$,

$$\frac{\cos(\pi/3)}{\cot(\pi/3)} = \frac{1/2}{1/\sqrt{3}} = \frac{\sqrt{3}}{2}.$$

Therefore a range of $\{0\}$ is not an equivalent representation, a harmless simplification, or a branch-choice artifact: it omits a directly attained real value. Since the original quotient removes the points where $\sin x = \pm 1$, the endpoint values ± 1 are approached but not attained, giving the range $(-1, 1)$.

1.5 Source-Level Diagnosis

The diagnosis identifies two coupled steps in the calculus utilities. The relevant files are `sympy/calculus/singularities.py` and `sympy/calculus/util.py`. The traced public call path is through `continuous_domain` and `function_range`. The former applies function-specific constraints and subtracts the singularities of the expression; the latter computes a period, reduces the infinite real domain to one period, calls `continuous_domain`, samples interval endpoints, solves $f'(x) = 0$, and builds the range from those values.

The first faulty step is in `sympy/calculus/singularities.py`, where the singularity finder rewrites trigonometric functions before searching for negative powers:

```
e = expression.rewrite([sec, csc, cot, tan], cos)
...
for i in e.atoms(Pow):
    if i.exp.is_negative:
        sings += solveset(i.base, symbol, domain)
```

For $\cos(x)/\cot(x)$, rewriting $\cot(x)$ as $\cos(x)/\sin(x)$ turns the quotient into an expression equivalent to $\sin(x)$ only away from the discarded denominator constraints. The condition $\cot(x) = 0$, which excludes $x = \pi/2 + k\pi$, is lost, so `singularities` returns `EmptySet` and `continuous_domain` reports all reals.

The second faulty step is in `sympy/calculus/util.py`. After the missed singularities, `function_range` treats the periodic domain as a closed interval such as $[0, 2\pi]$. Endpoint evaluation gives 0 at both ends. The recorded diagnosis notes that solving $f'(x) = 0$ on that interval returns no critical points for the unsimplified expression. The range is then constructed as the singleton $\{0\}$, missing interior values such as $\sqrt{3}/2$.

A plausible fix direction supported by the diagnosis is to preserve denominator constraints before applying trigonometric rewrites in `singularities`, or to add explicit zero and pole sets for tan, cot, sec, and csc. The diagnosis also notes that `function_range` would be more robust if it treated undefined points inside a periodic interval as domain cuts before endpoint and critical-value sampling.

1.6 Additional Failing Instances

The same failure appears for the representative expression and five shifted instantiations:

$$f_a(x) = \frac{\cos(x+a)}{\cot(x+a)}, \qquad a \in \{0, 1, 2, 3, 4, 5\}.$$

For each shift, the sample point $x = \pi/3 - a$ is defined and has value $\sqrt{3}/2$, while $x = \pi/2 - a$ is undefined. The same reasoning applies after the shift: the quotient agrees with $\sin(x + a)$ only on the original quotient's punctured domain, so a singleton range $\{0\}$ and a full-real domain are both invalid.

Input / Expression	**SymPy behavior**	**Expected behavior**
$\cos(x)/\cot(x)$	Reports domain $\mathbb{R}$ and range $\{0\}$; omits $\sqrt{3}/2$	Exclude $x = k\pi/2$; include $\sqrt{3}/2$; range $(-1, 1)$
$\cos(x+1)/\cot(x+1)$	Same collapse at $x = \pi/3 - 1$	Exclude $x = k\pi/2 - 1$; include $\sqrt{3}/2$; range $(-1, 1)$
$\cos(x+2)/\cot(x+2)$	Same collapse at $x = \pi/3 - 2$	Exclude $x = k\pi/2 - 2$; include $\sqrt{3}/2$; range $(-1, 1)$
$\cos(x+3)/\cot(x+3)$	Same collapse at $x = \pi/3 - 3$	Exclude $x = k\pi/2 - 3$; include $\sqrt{3}/2$; range $(-1, 1)$
$\cos(x+4)/\cot(x+4)$	Same collapse at $x = \pi/3 - 4$	Exclude $x = k\pi/2 - 4$; include $\sqrt{3}/2$; range $(-1, 1)$
$\cos(x+5)/\cot(x+5)$	Same collapse at $x = \pi/3 - 5$	Exclude $x = k\pi/2 - 5$; include $\sqrt{3}/2$; range $(-1, 1)$

1.7 Related Issues Assessment

The bug-finding harness performed a best-effort deduplication check. The closest public material found was SymPy's calculus documentation for `continuous_domain` and `function_range`; no public report was identified for `function_range(cos(x)/cot(x), x, S.Reals)` returning $\{0\}$, for `continuous_domain(cos(x)/cot(x), x, S.Reals)` returning all reals, or for this exact reciprocal-trigonometric singularity loss. The deduplication verdict is therefore a new failure mode with no public precedent. The bug remains individually actionable because it supplies a minimal reproducible counterexample and a focused regression test for the calculus domain/range utilities.

1.8 Proposed SymPy Regression Test

```python
import pytest
from sympy import S, cos, cot, pi, sqrt, symbols
from sympy.calculus.util import continuous_domain, function_range


@pytest.mark.parametrize("a", [0, 1, 2, 3, 4, 5])
def test_function_range_cos_over_cot_contains_defined_sample(a):
    x = symbols("x")
    expr = cos(x + a)/cot(x + a)

    assert expr.subs(x, pi/3 - a) == sqrt(3)/2
    assert expr.subs(x, pi/2 - a) is S.NaN
    assert continuous_domain(expr, x, S.Reals) != S.Reals
    assert function_range(expr, x, S.Reals).contains(sqrt(3)/2) is S.true
```